

# Toward Understanding the Effects of Visual and Tactile Stimuli to Reduce the Sensation of Movement with XR Mobility Platform

Taishi Sawabe\*

Masayuki Kanbara†

Yuichiro Fujimoto‡

Hirokazu Kato§

Nara Institute of Science and Technology

## ABSTRACT

This paper investigates a reduction method for passenger's movement sensation with the XR mobility platform mounted on an autonomous vehicle to improve passenger comfort during auto-driving. We investigate a reduction method that controls passenger's sense of movement by controlling visual and tactile perception using a multimodal XR mobility platform which consists of an immersive display and a motion platform with a tilting seat. The result of 30 subjects shows the sense of movement perceived by the passenger was reduced significantly when both visual acceleration and tactile acceleration control method was activated inside a moving autonomous vehicle.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality;

## 1 INTRODUCTION

In the field of autonomous vehicle research, safety issues and efficiency problems are always focused to realize the safe auto-driving world, however, considering passenger comfort is also important in autonomous vehicles in terms of widespread in human society [5]. Although technology to diminish the real acceleration generated by auto-driving is essential to realize comfortable autonomous vehicles. There is research controlling the passenger's visual perception and tactile perception of movement sense [1, 2, 4]. They show the effect of using HMDs or fixed displays to induce a strong presence by visual stimuli. Morita et al, proposed a telepresence system presenting vehicle-ride sensation in real-time by using a personal vehicle for a local driver, and a rotary chair with vestibular haptics feedback [3]. However, they only move in one angle of rotation movement on the visual and tactile perception. It is not enough angles to control a passenger's perception inside real autonomous vehicles.

## 2 REDUCTION METHOD OF SENSE OF MOVEMENT

The conventional method of pseudo-presenting acceleration has been to tilt the seat and use part of the vertical gravity to pseudo-present horizontal acceleration other than gravity. In the example of acceleration, the user receives a backward acceleration stimulus due to acceleration, so the seat is tilted to match the direction of the combined force of gravity and acceleration with the real gravity. This creates an illusion of backward acceleration for the user, who is in a visually immersive environment and perceives that he is horizontal from his vision. The important thing to remember is that the user should not perceive the true horizontal. In this study, the seat is tilted in the opposite direction, and the passenger perceives

the actual generated acceleration stimulus as a vertical direction, thereby reducing the acceleration stimulus of parallel movement.

Although humans also perceive a sense of movement through auditory sensations, subjects were asked to wear noise-cancelling earphones with music playing. We use an omnidirectional camera (Insta360 One X2) to capture a full-dome image for the visual cue of acceleration. Subjects view the image through an HMD (Oculus Rift S) as an immersive display so that they cannot perceive the actual horizontal.

## 3 EXPERIMENT

In this experiment, we evaluate 4 conditions A to D of acceleration stimuli to the sense of movement perceived by passengers in an automated vehicle: with and without acceleration stimuli from visual, and with and without acceleration stimuli from force perception. A where the passenger is riding in a car, as usual, to detect acceleration stimuli from visual and force sensations. In the proposed method of D, there is no acceleration stimulus from both visual and force senses, so the subject can be assumed to feel the least sense of movement. B and C are conditions without force and visual presentation, respectively, where inconsistency occurs between visual and force senses.

30 subjects (27 males and 3 females, mean age 27 (SD=8.8), age range 22 - 62) were tested. Subjects ride the XR mobility platform and subjectively evaluate their impressions in 4 conditions is shown in Figure 1. The subject sits in a tilt-controlled seat with a seatbelt and wears an HMD. At the same time, they wear noise-cancelling headphones with a loud enough sound to prevent perceiving the sense of movement from the sound. In the experiment, the automated vehicle automatically drives along a preset route around a 300 m oval course. The vehicle rotates in a clockwise direction and runs at a maximum speed of 30 km per hour. First of all, subjects were to ride in condition A as a training run, and were instructed to use the sense of movement they felt from this run as the standard for the amount of movement. After that, four conditions A-D were randomly ridden by each subject. After each ride, subjects were asked to answer questions about their sense of movement and the consistency between visual movement and body movement.

## 4 RESULTS

### 4.1 Sense of movement

The result of average values of the sense of movement with the 7 points Likert Scale, for 1 is no movement, and 7 is a lot of movement is shown in Figure 2. The value of proposed method D is 3.6 with a standard deviation of 1.9. For a test of significance, first, we use the Shapiro-Wilk test to see a test of normality in frequentist statistics. As the result, there is non-normality for each group. Next, we use the Friedman test for a non-parametric statistical test, that shows the existence of significance with  $p < .005$ . Therefore, we use the Bonferroni correction as a method to counteract the problem of multiple comparisons. As the result, there are significance in 2 conditions with  $p < .005$ , B-D, and A-D.

\*e-mail:t.sawabe@is.naist.jp

†e-mail:kanbara@is.naist.jp

‡e-mail:yfujimoto@is.naist.jp

§e-mail:kato@is.naist.jp

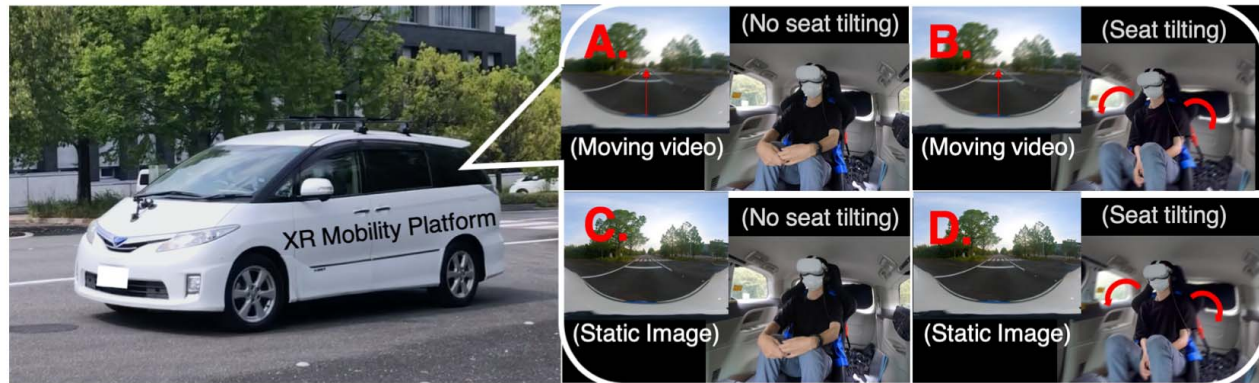


Figure 1: XR mobility platform consists of immersive displays and a motion platform inside an autonomous vehicle with four conditions.

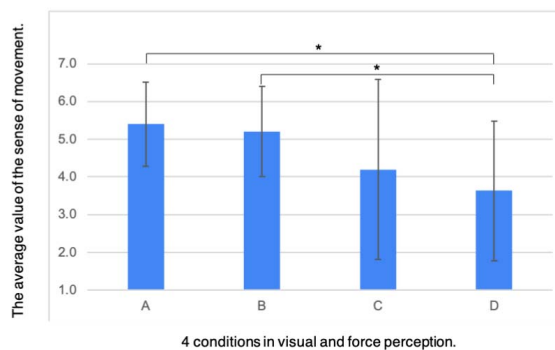


Figure 2: Sense of movement under four conditions.

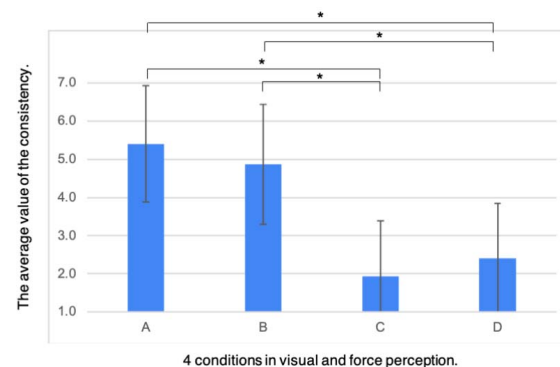


Figure 3: Sense of consistency under four conditions.

## 4.2 Consistency

The result of the average consistency with the 7 points Likert Scale, for 1 is very mismatching, and 7 is very matching is shown in Figure 3. The value of proposed method D is 2.4 with a standard deviation of 1.5. For a test of significance, first, we use the Shapiro-Wilk test to see a test of normality in frequentist statistics. As the result, there is non-normality for each group. Furthermore, we use the Friedman test for a non-parametric statistical test, that shows the existence of significance with  $p < .005$ . Therefore, we use the Bonferroni correction as a method to counteract the problem of multiple comparisons. As the result, there is significance in 4 conditions with  $p < .005$ , C-B, C-A, D-B, and D-A.

## 4.3 Discussion

Result of sense of movement, proposed method of method D which is the no acceleration stimulus from both visual and force senses showed less sense of movement than other methods. Results suggests that by using visual and tactile stimuli could control passenger's sense of movement inside autonomous vehicle environment.

## 5 CONCLUSION

This paper proposed a method to reduce the sense of movement by using visual and tactile stimuli with a multi-modal XR system mounted on an autonomous vehicle. From the results of 30 subjects, it was confirmed that the sense of movement was reduced significantly when both visual and force acceleration stimuli were not presented by controlling passenger's sensation. This method enables

passengers to perceive the vehicle as if they are not moving, even though they are in an automated driving environment.

## ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Number 21H03484.

## REFERENCES

- [1] T. M. Benz, B. Riedl, and L. L. Chuang. Projection displays induce less simulator sickness than head-mounted displays in a real vehicle driving simulator. *International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, pp. 379–387, 2019.
- [2] R. Kodama, M. Koge, S. Taguchi, and H. Kajimoto. Coms-vr: Mobile virtual reality entertainment system using electric car and head-mounted display. *2017 IEEE Symposium on 3D User Interfaces (3DUI)*, pp. 130–133, 2017.
- [3] T. Morita, T. V. Yem, Amemiya, and Y. Ikei. Vehicle-ride sensation sharing for immersive remote collaboration with vestibular haptic chair to reduce vr sickness. *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE 2019)*, pp. 1–2, 2019.
- [4] T. Sawabe, M. Kanbara, and N. Hagita. Diminished reality for acceleration stimulus: Motion sickness reduction withvection for autonomous driving. *IEEE Virtual Reality*, pp. 277–278, 2017.
- [5] T. Sawabe, M. Kanbara, and N. Hagita. Comfortable intelligence for evaluating passenger characteristics in autonomous wheelchairs. *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, pp. 1308–1316, 2018.